

Education

University of Utah
B.S. Computer Science

Salt Lake City, UT
August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies
Associate Software Engineer

Dallas, TX
June 2025 – Present

- Developed real-time signal processing algorithms in C++ for high-frequency defense sensor data streams, applying multithreading and concurrency techniques to meet strict sub-millisecond latency and throughput constraints in production systems.
- Built containerized microservices for high-frequency data ingestion and storage on embedded Linux, profiling and optimizing critical code paths to maximize throughput under resource constraints.
- Automated CI/CD pipelines with static analysis and integration testing, maintaining deployment reliability across multiple production services running on live hardware.

Doxy.me
Software Engineer

Charleston, SC
August 2024 – May 2025

- Built a real-time mental health inference system for Doxy.me's telehealth platform, fine-tuning Llama models to predict depression risk, anxiety, and mood indicators from tone, voice, and facial cues captured over live WebRTC sessions, served via SageMaker and Fargate.
- Improved prediction accuracy of the mental health inference system by applying RL feedback loops over structured clinical signals and physician-validated outcomes from live video consultations.
- Eliminated environment drift and enabled zero-downtime releases by standardizing CI/CD configuration with YAML-based pipelines across the ML stack.

University of Utah
Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT
May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Ray Tracing Engine: Created an interactive WebGL-based ray tracing engine in JavaScript, featuring realistic reflections, dynamic lighting, and customizable environment maps. Implemented shaders and user controls for rendering techniques, scene adjustments, and interactivity.